

FactoryDecoder: Expertise-Free Digital Twin Generation and Modification Tool

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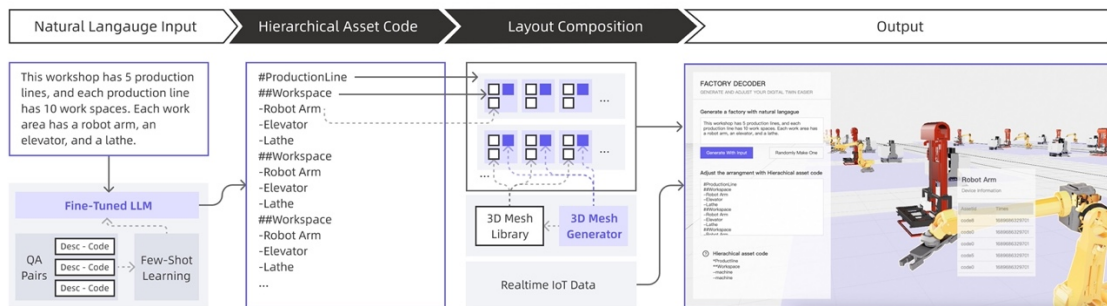


Figure 1: System overview of FactoryDecoder.

ABSTRACT

Digital twins are essential visualization applications in the manufacturing field. However, their development requires specialized 3D engineers, which often complicates modifications during the operation stage. To address this complexity, we introduce FactoryDecoder, a development tool for non-expert users in 3D engineering to generate and modify digital twins using natural language inputs. FactoryDecoder converts users' descriptions of production line into hierarchical asset codes, facilitating the automated layout and simplified modification of digital twins. Furthermore, if the system finds that the 3D asset library lacks appropriate device representations, it will automatically use a 3D mesh generator to create new ones. We evaluate the performance of large language models (LLMs) to optimize FactoryDecoder's capabilities. Preliminary user studies highlight FactoryDecoder's effectiveness.

CCS CONCEPTS

• Human-centered computing; • Visualization; • Visualization application domains;

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KEYWORDS

Generative artificial intelligence, Large language model, Digital twin, Data visualization

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1 INTRODUCTION

A digital twin is a virtual representation that mirrors a physical entity or process in manufacture field. Currently the creation of a digital twin scenario often necessitates expert engineers who are proficient in complex 3D software for device representation modeling and scenario building. This requirement makes modifications challenging for non-expert users in 3D development, such as production managers or operational staffs.

The integration of generative artificial intelligence (GenAI) in digital twin development offers potential solutions. Studies on the generative construction of digital twins range from generating 3D scenes [Long et al., 2023, Sun et al., 2023], single 3D models [Jun and Nichol, 2023, Li et al., 2023], to data structures utilized in digital twins [Banerjee et al., 2017]. However, these tools mostly focus on the generation stage with insufficient consideration for precisely modifying generated objects in production, lacking a solution for non-expert users.

To address this complexity, we introduce FactoryDecoder, a development tool for non-expert users in 3D engineering to generate



Figure 2: Hierarchical asset code definition.

and modify digital twins using natural language inputs. FactoryDecoder translates descriptions of production lines into hierarchical asset codes. These codes will guide organizing scenario layouts and device representations positioning. Moreover, the system will automatically engage a 3D mesh generator [Li et al., 2023] to generate new representations if it detects a lack of suitable 3D representations. Users can interact with the device 3D representations, monitoring real-time device data, and adjust the layout by directly modifying the hierarchical asset codes. By leveraging GenAI, FactoryDecoder can significantly reduce digital twin development cost.

2 SYSTEM DESIGN

To enhance the explainability and minimize inaccuracies in GenAI output, we introduce a markup language-inspired hierarchical asset code system as shown in Figure 2. Based on the hierarchical asset code, we build FactoryDecoder, a digital twin development tool enhanced by GenAI. The system consists of three modules: an internet of things (IoT) data platform, a code generator, and a layout compositor. This system can transform textual descriptions of a production line into hierarchical asset codes, which subsequently serve as a blueprint for generating corresponding 3D digital twin. If it recognizes a deficiency of suitable 3D representations, the system will automatically invoke a 3D mesh generator to create new representations. Furthermore, the production line generated is connected to real-time device instances from the IoT data platform. For modification, users can directly modify the hierarchical asset code like writing markdown to adjust the layout or replace device instances.

3 EVALUATION

3.1 Few-shot learning evaluation

To evaluate the applicability of LLMs in generating correct hierarchical asset codes, we conducted a comparative experiment. We prompted six models with 3-shot, 5-shot, and 10-shot from the training set and subsequently tested them on the entire test set. We captured the responses and assessed them via a 5-point Likert scale for correctness, merging these evaluations with complexity weights to compute a weighted average. The experimental outcomes is shown in Figure 3. Notably, Qwen2’s performance improved significantly with CoT.

3.2 Preliminary User Test

We carried out preliminary user tests with three manufacturing field experts (all males; $n=3$, mean age=38). They were introduced to the system followed by semi-structured interviews. All participants acknowledged the interaction positively. They appreciated FactoryDecoder for its flexibility and speed, which far surpass those

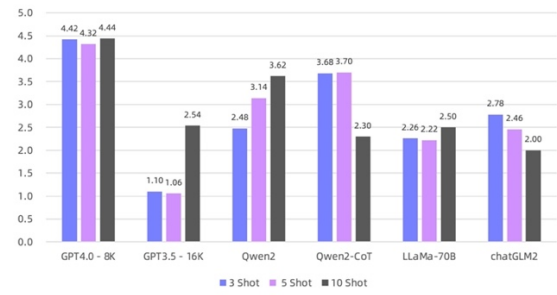


Figure 3: Results of few-shot learning evaluation.

of drag-and-drop methods. Furthermore, they recognized the ability to modify the codes and observe the changes instantaneously as a significant advantage. For suggestions, participants noted that current hierarchical asset code could still be improved in reflecting finer real-world details, such as pipeline connections between devices.

4 CONCLUSION

In this study, we introduce FactoryDecoder, a novel system designed to generate and modify 3D digital twins of factory production lines using natural language input and hierarchical asset code. Preliminary user studies point out that code-based GenAI provides more potentials in explainability as well as intervention ability.

Currently, FactoryDecoder demonstrates only three levels of hierarchy and lacks the ability to connect pipelines, a capability that is crucial in SCADA systems. Our future work would focus on defining a richer set of hierarchical asset codes yet maintains simplicity and readability. Building upon this, we could do segmentation and codification with factory production line photos, which can then be utilized to re-render scenes.

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